NELANI GUNAWARDHANA



B.Sc. (Hons) Microbiology Candidate | Natural Product & Computational Research Enthusiast

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PERSONAL PROFILE

Analytical and high-achieving Biosciences undergraduate combining a top-tier academic record in microbiology with wet-lab, bioinformatics, and molecular modeling research. Demonstrates exceptional leadership and project management through key university club appointments, international fellowship experience, and high-impact community development initiatives. Recognized for driving scientific literacy, effective public communication, and cross-functional team execution.

EDUCATION & ACADEMIC RECORD

University of Sri Jayewardenepura — B.Sc. (Honours) in Microbiology **Current GPA: 3.84 / 4.00**

Faculty of Applied Sciences | Department of Botany

- **Academic Rank:** Ranked Top in Batch for Special Degree selection with a 4.00 GPA in both Microbiology and Zoology.
- **Subject Breakdown:** Microbiology GPA: 3.97 | Zoology GPA: 4.00 | Chemistry 3.33 Selected for Special Degree track in all 03 subjects, Microbiology, Zoology and Chemistry
- **Key Honors:** Gold Medalist – IOBSL Interuniversity Biology Quiz (2024; 1 st Place across Sri Lanka).

RESEARCH PROJECTS

- **Honours Thesis (Ongoing) — In Vitro & In Silico Colorectal Cancer Bioactives**

Integrated evaluation of anticancer bioactivity and gut microbiota synergism of *Acrostichum aureum* L. leaf/fiddlehead compounds against human colorectal cancer targets utilizing molecular docking and wet-lab assays.

- Isolation and characterization of seed-borne endophytes from *Glycine max* and *Zea mays* for Indole-3-Acetic Acid (IAA) and Gibberellic Acid production. **2025**
- Isolated thermophilic bacteria, algae, and cyanobacteria from Madunagala Hot Springs. **2025**
- Evaluated biocontrol activity of *Trichoderma sp.* against rhizospheric fungal pathogens in *Abelmoschus esculentus*. **2025**

TECHNICAL SKILLS & BIOINFORMATICS

Laboratory & Wet-Lab: Aseptic Technique, Microbial Culture Maintenance, DNA Extraction, PCR, UV-Vis Spectroscopy, Brine Shrimp Cytotoxicity Bioassays, Laminar Flow, Fluorescence Microscopy.

In Silico & Bioinformatics: AutoDock, AutoDock Vina, PyRx, PyMOL, GROMACS (MD Simulations), UCSF Chimera, BLAST, ScanProsite, antiSMASH, Prokka.

Data Analysis & Computing: RStudio (ANOVA, Linear Regression, Experimental Design), Statistical Modeling.

EXPERIENCE & INTERNSHIPS

National Institute of Fundamental Studies (NIFS), Kandy — Research Intern **Mar 2026 – Apr 2026**

Microbiology & Soil Ecosystems Division

Executed culture collection maintenance, aseptic subculturing, UV-Vis spectroscopy, phycocyanin pigment trials, fluorescence microscopy, and brine shrimp lethality bioassays.

Pan Asia Bank PLC — Corporate Intern **2022 – 2023**

Supported corporate operations and customer relationship management; recognized at the branch awards.

FELLOWSHIPS & DIPLOMATIC ENGAGEMENTS

Young South Asian Leaders Initiative (YSALI) Fellow — U.S. Dept. of State **May – Jun 2026**

Selected Sri Lankan representative for an immersive exchange organized by UNO, East-West Center, focusing on STEM innovation, entrepreneurship, and youth leadership held in person in USA.

Invited Speaker & Panelist — American Spaces & Diplomatic Roundtables **2023 – 2026**

Panelist at Youth Forum Leadership Exchange Summit (Batticaloa, 2026).

Participated in high-level diplomatic dialogue with U.S. Ambassador Eric Meyer and Regional Public Engagement Specialists.

KEY LEADERSHIP ROLES

- **Secretary** : J'PURA VOICE Official Media Unit (2025/26)
- **Assistant Secretar** : Rotaract Club of USJ (2024/25)
- **Divisional Head** : Programming Division, J'PURA VOICE (2024/25)
- **Task Master** : Content Writing J'PURA VOICE (2023/24)
- **Assistant Director** :International Service Avenue Rotaract Club of USJ (2023/24)
- **Finance Manager** :DesignXcellence 1.0 Programming Division, J'PURA VOICE (2023/24)
- **Project Chairperson** :ARTIFEX 1.0 Programming Division, J'PURA VOICE (2023/24)
- **Project Chairperson** : A Memorable December, Club Service Avenue- Rotaract Club of USJ (2023/24)
- **Project Chairperson** : Eco Bricks, Environment Service Avenue - Rotaract Club of USJ (2023/24)
- **Advanced First Aider** : Sri Lanka Red Cross Society (2019–Present)

KEY HONORS & AWARDS

- **Gold Medalist** : IOBSL Inter-University Biology Quiz (2024)
- **Volunteer Appreciation Award** : Sri Lanka Red Cross Society (2024)
- **Best Leading Figure** : Rotaract Club USJ (2023/24)
- **Most Outstanding Programmer** : J'PURA VOICE (2023/24)
- **Voice of the Month (October)** : J'PURA VOICE (2023/24)
- **Best Member** : International Service Avenue, Rotaract USJ (2023/24)
- **Leo of the Month** : Leo Club 306 C2 District, of University of Sri Jayewardenepura (2023/24)
- **Most Outstanding Project of the Club Service Avenue, A Memorable December 5.0** : Rotaract Club of USJ (2023 / 2024)
- **Most Innovative Project of the Year, Eco Bricks** : Rotaract Club of USJ (2023 / 2024)

DECLARATION

I hereby declare that the information provided above is true, correct, and complete to the best of my knowledge.

REFERENCES

1. Dr. Isuru Kariyawasam
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